

# Pranav Chauhan

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## SUMMARY

Software engineer specializing in **full-stack development and LLM pipeline engineering**. Built a production-grade RAG system (AWS Transcribe -> LangChain -> pgvector -> Gemini), implemented **OAuth 2.0 Authorization Code Flow** from scratch with cryptographic token security, and architected a **distributed autonomous agent** with a 7-state machine and heartbeat-based crash recovery. Comfortable working across the entire stack, including React frontends, PostgreSQL schema design, agentic systems, and AI infrastructure.

## TECHNICAL SKILLS

|                        |                                                                   |
|------------------------|-------------------------------------------------------------------|
| <b>Languages:</b>      | C++, JavaScript (ES6+), TypeScript, Python, SQL                   |
| <b>Frontend:</b>       | React.js, Next.js, Tailwind CSS, Vite                             |
| <b>Backend:</b>        | Node.js, Express.js, Prisma, Drizzle ORM, REST APIs, JWT          |
| <b>DB &amp; Cloud:</b> | PostgreSQL (pgvector), MongoDB, SQLite, AWS S3, AWS Transcribe    |
| <b>AI &amp; Tools:</b> | Gemini, LangChain, pgvector embeddings, Supmemory, Git, OAuth 2.0 |

## KEY PROJECTS

**ByteLearn - AI-Powered Learning Platform** | *React 19, Node.js, PostgreSQL, AWS, Gemini, LangChain*

GitHub: [github.com/Pranav04027/bytetrainAI](https://github.com/Pranav04027/bytetrainAI) Live Demo: [bytetrainai.pranavch.tech](https://bytetrainai.pranavch.tech)

- **Built a 4-stage AI pipeline** using AWS Transcribe -> LangChain chunking (500-char segments, 50-char overlap) -> 768-dim Gemini embeddings -> pgvector similarity search, powering transcript-aware Q&A streamed to the client via SSE.
- **Shipped 7 AI-powered features**, including RAG Q&A over video transcripts, AI-generated MCQ quizzes from lesson content, a **learner memory system** (Supmemory + Gemini) that persists knowledge and struggle signals to personalize future responses, and engagement-based video recommendations.
- Architected a **19-model PostgreSQL schema** and 60+ REST endpoints across 16 route files, with modular Express controllers, Prisma ORM, and layered middleware for validation, auth, and error handling.
- **Implemented JWT-based auth with RBAC** - access/refresh token pair with silent refresh via Axios interceptor, refresh token rotation, role-enforced middleware on instructor endpoints, and 16 protected frontend routes with role-conditional rendering.
- **Secured all media delivery via time-limited S3 presigned URLs** - videos are never publicly accessible; upload and playback URLs expire after use, preventing unauthorized sharing.
- Built a **background transcription polling loop** (30s interval) that detects AWS Transcribe job completion, fetches the transcript from S3, saves it to the DB, and automatically triggers the embedding pipeline - connecting upload to semantic search with zero manual intervention.

**AetherID - OAuth 2.0 Identity Provider** | *Next.js, MongoDB, TypeScript, JWT*

GitHub: [github.com/Pranav04027/AetherID](https://github.com/Pranav04027/AetherID) Live Demo: [aetherid.pranavch.tech](https://aetherid.pranavch.tech)

- **Implemented the full OAuth 2.0 Authorization Code Flow** from scratch - client registration, redirect URI validation, one-time auth code issuance (SHA-256 hashed, 7-min TTL via MongoDB TTL index), token exchange, and an OIDC-compliant /userinfo endpoint.
- **Engineered a cryptographic token security model** - SHA-256 hashing applied to auth codes, refresh tokens, and verify/reset tokens; raw secrets never persisted, making a database breach non-exploitable.
- **Built refresh token rotation with aggressive reuse detection** - 15-min access tokens paired with 7-day refresh tokens; if a rotated token is replayed, all sessions for that user are immediately invalidated to prevent token theft attacks.
- **Developed a dual-layer auth system**: HttpOnly cookie-based sessions for the internal dashboard and rotating JWT pairs for third-party OAuth clients, with OAuth **state parameter** support for CSRF protection on redirects.

**Beav - Autonomous AI Engineer Orchestrator** | *TypeScript, Node.js, SQLite, Drizzle ORM, Codex API*

GitHub: [github.com/Pranav04027/Beav](https://github.com/Pranav04027/Beav)

- **Architected a 5-package TypeScript monorepo** (tracker, orchestrator, worker, core, CLI) to autonomously detect GitHub issues, clone isolated workspaces, run an LLM coding agent via **JSON-RPC 2.0 over stdin/stdout**, and submit PRs - with a verification loop polling GitHub CI check conclusions via Octokit.
- **Designed a 7-state machine** (pending -> claimed -> running -> verifying -> done / crashed -> failed) that enforces valid transitions to prevent double-claiming while guaranteeing task integrity across up to **3 configurable concurrent workers**.
- **Built heartbeat-based crash detection** - workers emit IPC pulses every 15s; orchestrator force-kills any PID that remains silent for 45 seconds and requeues the task with **exponential backoff**, plus startup recovery that automatically requeues orphaned tasks from prior crashes.
- **Eliminated DB contention across parallel workers** using SQLite WAL mode and atomic state transitions, storing full task lifecycle and streaming worker logs (stdout/stderr/system) across 2 relational tables with foreign key integrity.
- **Secured agent execution with path-traversal-protected sandboxed workspaces** - each task receives a fresh shallow git clone; LLM-generated file operations are restricted to the workspace root, preventing escape to the host filesystem.

## EDUCATION

**Indian Institute of Information Technology Vadodara**  
B.Tech in Computer Science

Gandhinagar, Gujarat 382010  
2022-2026 | CPI: 7.64/10

**Maa Bharti Sr Sec School**  
HSC (Class XII)

Kota, Rajasthan 324001  
81.0%

## LEADERSHIP & ACHIEVEMENTS

Club Secretary, **Masquerade** - Led the team to **2nd place at IIT Gandhinagar** and **3rd place at IIT Kharagpur** in inter-college competitions.